INSTITUTO SUPERIOR TECNOLÓGICO QUITO

Excelencia en Educación Superior

2.1 Estandarizacion

python™

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In [5]:
```
from sklearn import preprocessing
import numpy as np
```

In [6]:
```
# Carga de datos.
X_train = np.array([[ 1., -1.,  2.],
                    [ 2.,  0.,  0.],
                    [ 0.,  1., -1.]])
```

In [7]:
```
# Limpieza de datos: estandarización.
standardizer = preprocessing.StandardScaler()

# Estandarizamos los datos de entrenamiento
X_train_std=standardizer.fit_transform(X_train)
print(X_train_std)
```

```
[[ 0.         -1.22474487  1.33630621]
 [ 1.22474487  0.         -0.26726124]
 [-1.22474487  1.22474487 -1.06904497]]
```

In [8]:
```
X_train_2 = np.array([[ 1., -1.,  2.],
                    [ 2.,  0.,  0.],
                    [ 0.,  1., -1.]])
```

In [9]:
```
# Aplicamos los métodos "fit" y "transform" por separado
f=preprocessing.StandardScaler()
f=f.fit(X_train)
```

```python
X_train_2_std = f.transform(X_train_2)
print(X_train_2_std)
# ???
```

```
[[ 0.         -1.22474487  1.33630621]
 [ 1.22474487  0.         -0.26726124]
 [-1.22474487  1.22474487 -1.06904497]]
```

In [11]:

```python
# Comprobar si las matrices estandarizadas son iguales
print(np.allclose(X_train_std, X_train_2_std))
```

```
True
```

GitHub

https://github.com/Erick-305/machine-learning.git